XYRA AI Studio

Private AI for EPCs

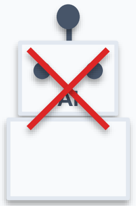
P&ID Intelligence | Piping MTO | Centralized Database | On-Premise

This is how engineering data moves today.

From the moment a P&ID is issued to the moment a deliverable is produced.

1. **P&ID Issued** — Drawing arrives
2. **Manual Extraction** — 2–4 hrs per sheet ✕ — 2–4 hrs of specialist time
3. **Spreadsheet Created** — System A
4. **Emails to Different Teams** — Last week's file ✕ — Wrong version sent
5. **Re-entered Into system** — System B - manual ✕ — Re-keyed manually
6. **Review Meeting** — Checks what was checked ✕ — Already checked last phase

The same instrument tag is entered 4–6 times across a single project.
Discipline-to-discipline data transfer happens through disconnected files - no traceability, no version control.



"We'll use ChatGPT."

- Doesn't know ISA-5.1 ✕
- Can't reliably read P&IDs ✕
- Client drawings can't go to a public AI ✕

30–50%

of your engineering budget
goes to clerical work.

Not engineering.

Where it goes

Breakdown of wasted engineering capacity on a typical EPC project:

Manual take-off

2–4 hrs per P&ID drawing, repeated across hundreds of drawings

Data re-entry

Same instrument tag entered 4–6× across systems by different teams

Discipline handoffs

Emailed spreadsheets, wrong versions, no audit trail between teams

Review & re-checking

Review phases repeat checks already done - no shared engineering record

There is a better way.

XYRA AI Studio

Private AI for EPC Drawing Deliverables

01

InstruMap

Extract instrument tags, classify with ISA-5.1, map to lines, generate Index / IO List / Verification Log.

02

Piping MTO

User-built component library + computer-vision detection across all drawings. EPC-style Excel output.

03

Centralized DB

PostgreSQL-backed engineering records - instruments, loops, piping, IO - shared across all disciplines.

04

Custom AI Solutions

Build custom engineering applications, automate project-specific workflows, fine-tune private AI models, and train systems using client standards, specifications, and historical project data.

xyra ai

This is how it works with XYRA.

One upload. Every tool. One structured output to all disciplines.

1 Upload P&ID	2 XYRA Extracts	3 Engineer Reviews	4 Download Outputs
Drag & drop PDF - any quality	ISA-5.1 decode Line mapping Service inference	Confidence flags QA notes One-click remove	Index · IO List MTO · Line List ZIP package
Drop PDFs no prep needed	LLM + OCR runs locally	Full audit trail QA flagged	Excel + ZIP review-ready

BEFORE XYRA

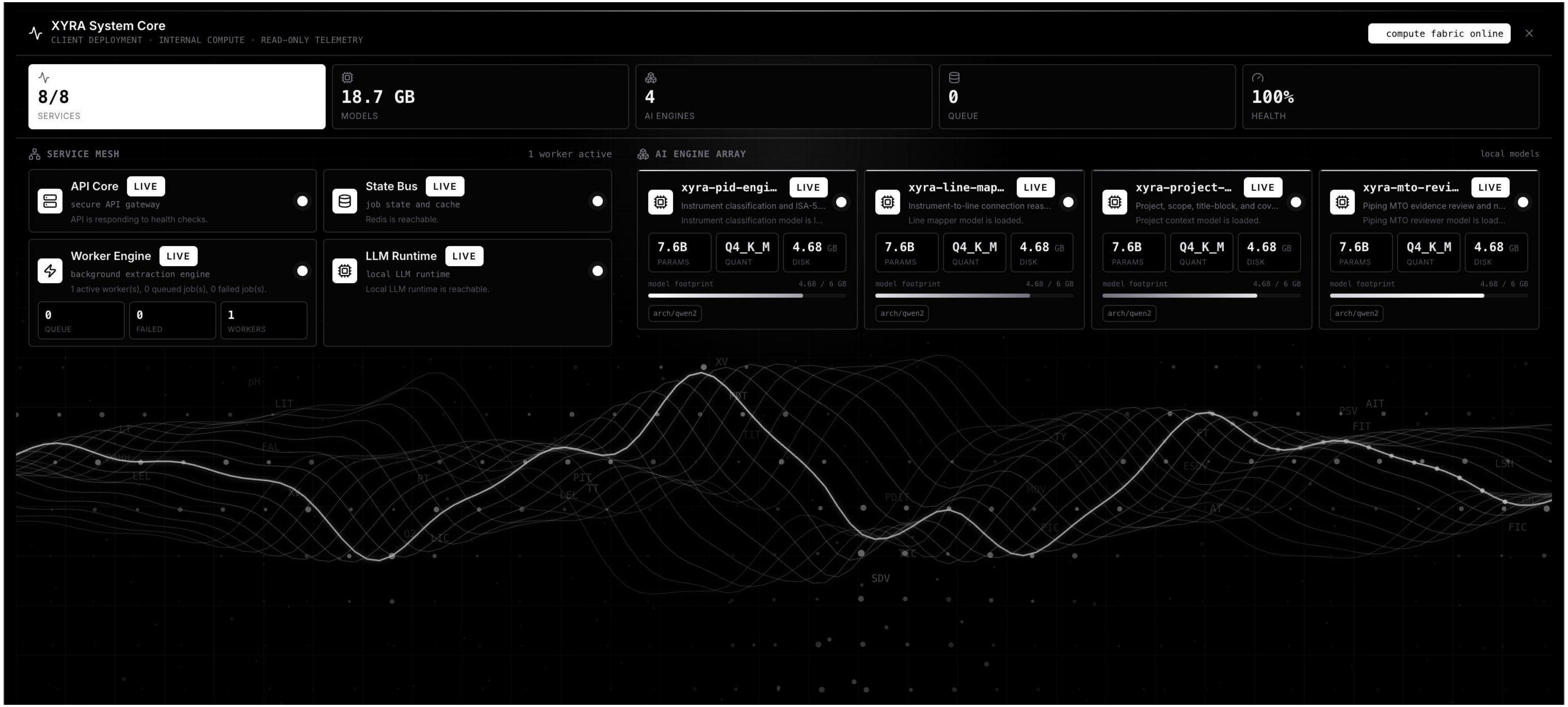
Manual take-off → 5 systems re-entered → email → wrong version → re-check

WITH XYRA

Upload → XYRA reads → engineer reviews → structured outputs to all disciplines

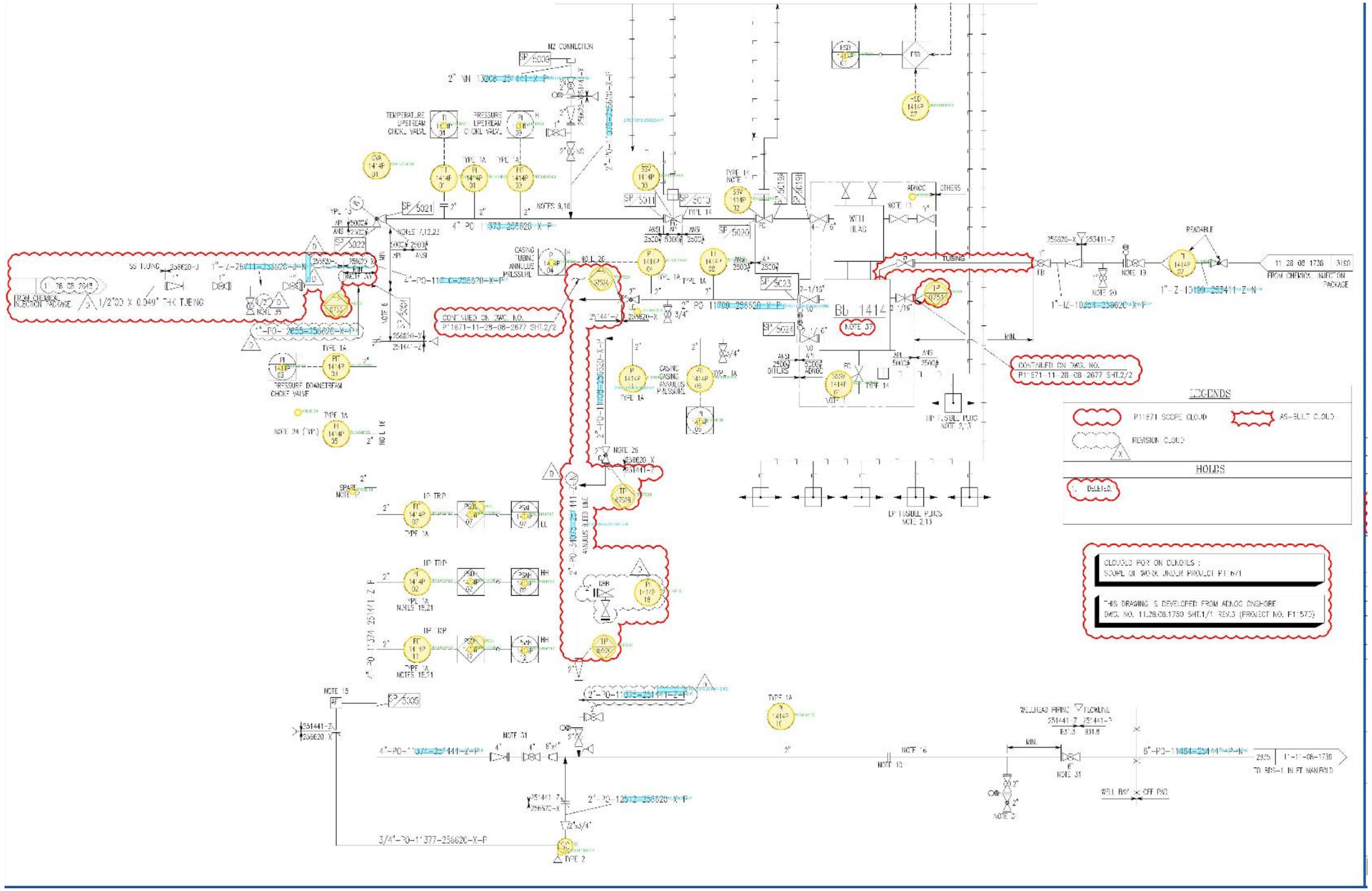
Compute fabric online.

8 / 8 services live · 4 custom AI engines · 18.7 GB models loaded · 100% health · 0 queue



XYRA reads the drawing and prepares check print

Every instrument tag identified, classified, and linked to its pipeline number.



Every instrument.

Instrument Index - tags · loops · ISA type · service description · IO type · P&ID reference · QA review flags

AutoSave · Instrument Index · Home Insert Draw Page Layout Formulas Data Review View Automate

A2 · FCV-9999P-12

	Tag No.	Loop No.	Instrument Description	Instrument Service	Service Confidenc	Service Basis	Instrument Ty	Suffi	P&ID No.	Line / Equip. Tag	Control Syste	IO Type	Signal Type
2	FCV-9999P-12	F-9999P-12	Flow Control Valve	Chemical injection line valve control	Medium	connected line 1-IZ-2684	FCV	12	P1234	1-IZ-26840-253411-Z-N	DCS	AO	4-20mA
3	FE-9999P-12	F-9999P-12	Flow Element	Flow element	Low	passive tag type only	FE	12	P1234		DCS	Soft Link	
4	FI-9999P-12	F-9999P-12	Flow Indicator	VG service line flow indication	Medium	connected line 6-VG-207	FI	12	P1234	6-VG-20715-013461-Y-N	DCS	Soft Link	
5	FIC-9999P-12	F-9999P-12	Flow Indicating Controller	Process flow control	Low	tag type only	FIC	12	P1234		DCS	Soft Link	
6	FIT-9999P-12	F-9999P-12	Flow Indicating Transmitter	VG service line flow	Medium	connected line 2-VG-207	FIT	12	P1234	2-VG-20718-013461-Y-N	DCS	AI	4-20mA + HART
7	FQI-9999P-12	F-9999P-12	Flow Totalizer Indicator	VG service line flow indication	Medium	connected line 6-VG-207	FQI	12	P1234	6-VG-20715-013461-Y-N	DCS	Soft Link	
8	FZT-9999P-12	F-9999P-12	Flow Element Position Transmitter	Control valve position feedback	High	FZT position feedback	FZT	12	P1234	1-IZ-26840-253411-Z-N	DCS	AI	4-20mA + HART
9	HS-9999P-03	H-9999P-03	Hand Switch	Operator hand switch	High	HS hand switch	HS	03	P1234	2-PG-24468-251482-X-N1	DCS	DI	24VDC (Dry Contact)
10	HS-9999P-04	H-9999P-04	Hand Switch	Operator hand switch	High	HS hand switch	HS	04	P1234	2-PG-24468-251482-X-N1	DCS	DI	24VDC (Dry Contact)
11	HS-9999P-06	H-9999P-06	Hand Switch	Operator hand switch	High	HS hand switch	HS	06	P1234	2-PG-24468-251482-X-N1	DCS	DI	24VDC (Dry Contact)
12	HIC-9999P-12	H-9999P-12	Hand Indicating Controller	Process instrument control	Low	tag type only	HIC	12	P1234		DCS	Soft Link	
13	MSAS-9999P-01	M-9999P-01	Moisture Switch Alarm Switch	PG service line instrument switch	Medium	connected line 2-PG-244	MSAS	01	P1234	2-PG-24468-251482-X-N1	DCS	DI	24VDC (Dry Contact)
14	PI-9999P-12	P-9999P-12	Pressure Indicator	VG service line pressure indication	Medium	connected line 6-VG-207	PI	12	P1234	6-VG-20715-013461-Y-N	DCS	Soft Link	
15	PIC-9999P-12	P-9999P-12	Pressure Indicating Controller	Process pressure control	Low	tag type only	PIC	12	P1234		DCS	Soft Link	
16	PIT-9999P-12	P-9999P-12	Pressure Indicating Transmitter	Pressure near SSV-9999P-03	High	same line/loop valve cor	PIT	12	P1234	2-PG-24464-251482-X-N	DCS	AI	4-20mA + HART
17	PY-9999P-12	P-9999P-12	Pressure Relay/Compute	Pressure signal conversion	Low	relay/converter tag type	PY	12	P1234		DCS	Soft Link	
18	PI-9999P-14	P-9999P-14	Pressure Indicator	PG service line pressure indication	Medium	connected line 2-PG-244	PI	14	P1234	2-PG-24461-251482-X-N	DCS	Soft Link	
19	PIT-9999P-14	P-9999P-14	Pressure Indicating Transmitter	PG service line pressure	Medium	connected line 2-PG-244	PIT	14	P1234	2-PG-24461-251482-X-N	DCS	AI	4-20mA + HART
20	PIT-9999P-15	P-9999P-15	Pressure Indicating Transmitter	Chemical injection line pressure	Medium	connected line 1-IZ-2684	PIT	15	P1234	1-IZ-26842-253461-Y-N	DCS	AI	4-20mA + HART
21	PSAH-9999P-15	P-9999P-15	Pressure Switch Alarm High	Chemical injection line pressure high switch	Medium	connected line 1-IZ-2684	PSAH	15	P1234	1-IZ-26842-253461-Y-N	DCS	DI	24VDC (Dry Contact)
22	PI-9999P-17	P-9999P-17	Pressure Indicator	PG service line pressure indication	Medium	connected line 2-PG-244	PI	17	P1234	2-PG-24468-251482-X-N1	DCS	Soft Link	
23	PIT-9999P-17	P-9999P-17	Pressure Indicating Transmitter	Pressure upstream of SSV-9999P-06	High	same line/loop valve cor	PIT	17	P1234	2-PG-24468-251482-X-N1	DCS	AI	4-20mA + HART
24	PI-9999P-18	P-9999P-18	Pressure Indicator	Pressure indication downstream of SSV-9999P-0	High	same line/loop valve cor	PI	18	P1234	2-VG-23007-253461-Y-N	DCS	Soft Link	
25	PIT-9999P-18	P-9999P-18	Pressure Indicating Transmitter	Pressure near SSV-9999P-06	High	same line/loop valve cor	PIT	18	P1234	2-PG-24468-251482-X-N1	DCS	AI	4-20mA + HART
26	PI-9999P-19	P-9999P-19	Pressure Indicator	NOTE service line pressure indication	Medium	connected line 2-NOTE-2	PI	19	P1234	2-NOTE-253461-Y	DCS	Soft Link	
27	PIT-9999P-19	P-9999P-19	Pressure Indicating Transmitter	Pressure downstream of SSSV-9999P-04	High	same line/loop valve cor	PIT	19	P1234	2-PG-24466-251482-X-N	DCS	AI	4-20mA + HART
28	PI-9999P-20	P-9999P-20	Pressure Gauge	Pressure indication near SSV-9999P-03	High	same line/loop valve cor	PI	20	P1234	2-PG-24464-251482-X-N			
29	PI-9999P-21	P-9999P-21	Pressure Gauge	Pressure indication near SSV-9999P-03	High	same line/loop valve cor	PI	21	P1234	2-PG-24464-251482-X-N			
30	PI-9999P-22	P-9999P-22	Pressure Gauge	Pressure indication upstream of SSV-9999P-06	High	same line/loop valve cor	PI	22	P1234	2-PG-24468-251482-X-N18			
31	RO-9999P-03	R-9999P-03	Radiation Orifice	Radiation Orifice	Low	description fallback	RO	03	P1234	2-NOTE-253461-Y	DCS	Soft Link	
32	SSV-9999P-03	S-9999P-03	Surface Safety Valve	Shutdown valve command	High	SSV shutdown valve	SSV	03	P1234	2-PG-24464-251482-X-N	SIS/ESD	DO	24VDC
33	SZSH-9999P-03	S-9999P-03	Safety Position Switch (High/Open)	VG service line position high switch	Medium	connected line 2-VG-230	SZSH	03	P1234	2-VG-23007-253461-Y-N	SIS/ESD	DI	24VDC (Dry Contact)
34	SZSL-9999P-03	S-9999P-03	Safety Position Switch (Low/Closed)	PG service line position low switch	Medium	connected line 2-PG-244	SZSL	03	P1234	2-PG-24466-251482-X-N	SIS/ESD	DI	24VDC (Dry Contact)

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Every line number.

Line List - pipe numbers · sizes · fluid codes · insulation spec · area codes · coordinates · P&ID source

Line List

	Line_Number	Pipe_Siz	Size_Unit	Fluid_Cod	Sequence_	Area_Cod	Insulation	Spec	P&ID_Filename	P&ID_Pa	Coordinates
2	2-PG-24466-251482-X-N	2	in	PG	24466	251482	X	N	pid (1).pdf	1	3975,3614
3	2-PG-24465-251482-X-N	2	in	PG	24465	251482	X	N	pid (1).pdf	1	5839,3181
4	2-VG-20718-013461-Y-N	2	in	VG	20718	013461	Y	N	pid (1).pdf	1	7410,3390
5	2-PG-24469-253461-Y-N	2	in	PG	24469	253461	Y	N	pid (1).pdf	1	6616,3182
6	2-PG-24461-251482-X-N	2	in	PG	24461	251482	X	N	pid (1).pdf	1	6535,5286
7	2-PG-24464-251482-X-N	2	in	PG	24464	251482	X	N	pid (1).pdf	1	5119,3289
8	2-VG-23007-253461-Y-N	2	in	VG	23007	253461	Y	N	pid (1).pdf	1	4617,2587
9	6-VG-20715-013461-Y-N	6	in	VG	20715	013461	Y	N	pid (1).pdf	1	6289,2212
10	2-PG-24468-251482-X-N	2	in	PG	24468	251482	X	N	pid (1).pdf	1	1929,3527
11	2-PO-34088-251441-Z-N	2	in	PO	34088	251441	Z	N	pid (1).pdf	1	660,3690
12	1-IZ-26842-253461-Y-N	1	in	IZ	26842	253461	Y	N	pid (1).pdf	1	5477,3695
13	1-IZ-26840-253411-Z-N	1	in	IZ	26840	253411	Z	N	pid (1).pdf	1	6351,3694
14	2-PO-34088-251441-Z-NDBB	2	in	PO	34088	251441	Z	NDBB	pid (1).pdf	1	916,3579
15	2-PG-24461-251482-X-NFE	2	in	PG	24461	251482	X	NFE	pid (1).pdf	1	6895,5289
16	2-D-11980	2	in	D	11980				pid (1).pdf	1	7231,3169
17	2-PG-24468-251482-X-N18	2	in	PG	24468	251482	X	N18	pid (1).pdf	1	2173,3517
18	2-NOTE-253461-Y	2	in	NOTE	253461	Y			pid (1).pdf	1	4830,2580
19	2-PO-34088-251441-Z-NLC	2	in	PO	34088	251441	Z	NLC	pid (1).pdf	1	860,3582

Prepare MTO in a Minute.

XYRA finds every instance - across every page, every rotation, with size extraction.

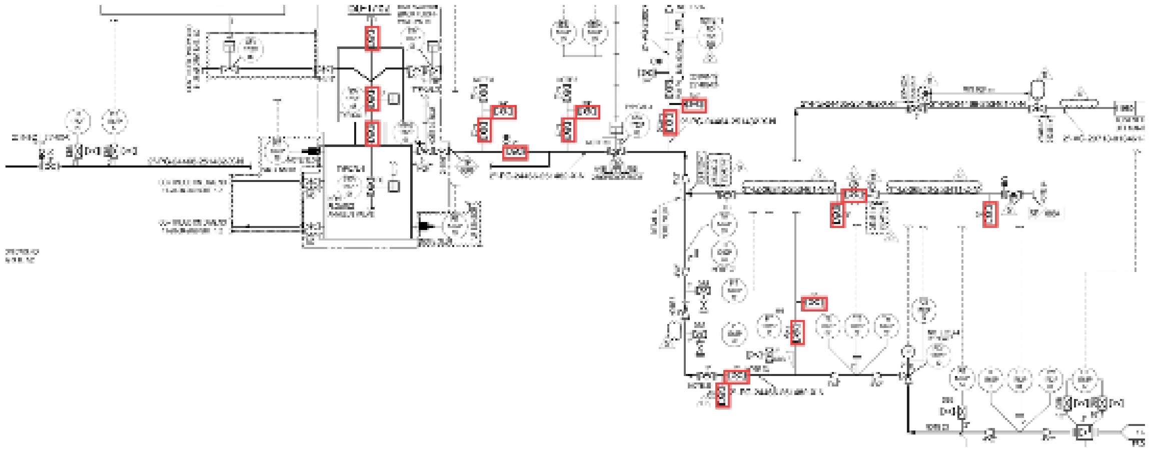
1 Select Component
Draw a box around any symbol

2 Save to Library
Auto-trimmed thumbnail stored

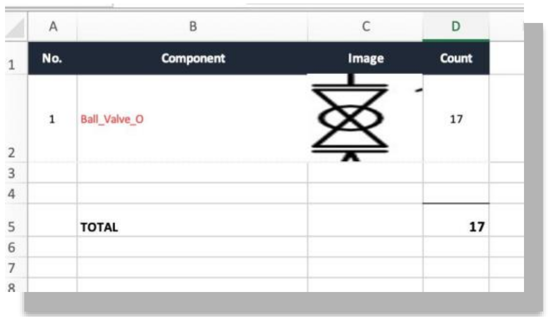
3 Run Detection
Computer vision across all pages

4 Review Results
Remove false positives

5 Export Package
MTO · Register QA · Excel ZIP



OUTPUT



No.	Component	Image	Count
1	Ball_Valve_O		17
	TOTAL		17

17 components detected

1 drawing processed

1 component type

Beyond Products – Engineering Automation Services

Engineering Tool Automation

- Custom DLLs and add-ins for Hexagon SmartPlant tools
- SmartPlant Instrumentation (SPI) automation
- Smart 3D (S3D) workflow automation
- Smart P&ID data extraction and validation
- Custom engineering productivity tools

Engineering Data Management

- Engineering database migration and cleansing
- Bulk data loading and validation
- Legacy data digitization
- Intelligent document processing

AI-Powered Engineering Solutions

- Private/on-premise AI assistants
- Custom AI models trained on engineering data
- Drawing intelligence and knowledge extraction
- Engineering document search and Q&A

Two weeks. One answer.

A structured pilot gives your team a clear decision: time saved, output quality, and deployment fit.

WEEK 1

Configure & Load

- Deploy on client server or test VM
- Load project legend + title block context
- Configure instrument model + MTO library
- Run first extraction pass - calibrate output

WEEK 2

Validate & Compare

- Process 20–50 representative P&IDs
- Compare XYRA output vs manual workflow
- Review Instrument Index, IO List, MTO
- Identify gaps - tune detection settings

DECISION PACK

Delivered at Close

- Coverage summary + exception log
- Time-saving estimate vs manual baseline
- Deployment checklist for production
- Monthly license proposal

Ready to see XYRA on your drawings?

We deploy inside your network, run XYRA on your drawing package,
and deliver structured outputs your team evaluates against the manual baseline.

Schedule a Pilot Run